of micropayments by the organizer (2, 3, and 4 in figure 7.8)--the only concern for the developer of MPC-Dapp is the digital signing of the micropayments. The organizer digitally signs the micropayments for later verification during the claim process.

Let’s examine the digital signing operation, which is a crucial feature of the MPC-Dapp and highly useful in blockchain-based digital networks of unknown participants.

DIGITAL SIGNING

What is digital signing? Consider a typical bank check, shown in figure 7.9. It has the amount, bank details, sender details, receiver details, check number, and signature of the sender (a unique signature authorizing the payment). The same figure shows a digital payment message of a payment channel with details mapped one-to-one with the traditional bank check.

Observe that bank details are replaced by the smart contract address and the check number by the unique nonce of the sender account. The micropayment message of the MPC could have all these data elements or a subset of them. The date on the traditional check is replaced by the timestamp of the transaction in the blockchain. The timestamp of a transaction is created at the time of recording, not at the time of creation of the message! You should be aware of these significant differences between bank checks and cryptopayment message. It is possible to pack more elements into a micropayment message to add to the robustness. You’ll use the minimum required data so as not to overload the blockchain.

Before the message is signed, it is hashed to a fixed size. (Chapter 5 discusses hashing in detail and provides examples.) Regardless of the number of items packed in a message or the